0 — Purpose and Scope

The EMBC Formulas document (Sections 5.2–5.3) establishes the Barkhausen feedback conditions for autonomous material response:

$$|G(\omega_0)| = |H_{\text{mat}}(\omega_0)| \cdot |H_{\text{fb}}(\omega_0)| \geq 1$$
$$\arg H_{\text{mat}}(\omega_0) - \omega_0 \tau_{\text{fb}} = 2\pi k, \quad k \in \mathbb{Z}$$

The Manufacturability Roadmap (Section 7.1) specifies the self-healing test target:

> *"≥ 80 % conductance recovery within 60 s"* (4-point probe, in-situ monitoring)

What was missing: A mechanistic explanation of *how* a micro-crack in the organosilicate matrix physically regenerates and restores electrical/dielectric function — the bridge between the mathematical feedback model and observable material behaviour.

This supplement provides that mechanism by adapting the Heliotherm X ReGen regenerative healing framework (ODE system) from the thermal-structural domain into the electro-chemical domain of the EMBC architecture.

1 — Structural Analogy: Heliotherm X ReGen ↔ EMBC

The Heliotherm X ReGen system provides the foundational healing ODE:

$$D(t) = D_0 - H(T, t)$$
$$H(T, t) = k_h \cdot S(t) \cdot e^{-m \cdot t}$$
$$\frac{dS}{dt} = R(T) - C(T, D(t))$$

where $S(t)$ is the available healing substance, $R(T)$ its regeneration rate, and $C(T, D)$ its consumption rate. The key insight is the **equilibrium condition** $R(T) \approx C(T, D) \Rightarrow S(t) = S_{\text{stable}}$, which yields practically unlimited healing cycles because the healing phase regenerates itself.

However, the two systems operate in fundamentally different domains:

Aspect	Heliotherm X ReGen	EMBC
Damage type	Structural crack (mechanical)	Dielectric/conductive discontinuity (EM)
Healing phase	ReGen-Phase (metastable ceramic, ε = 0.10)	Siloxane SiR_2O (mobile polymer chains)
Activation energy source	External temperature > 1 800 °C	Internal EM field at 13.56 MHz (self-activated)
Activation mechanism	Thermal diffusion	Dielectric loss heating at crack tip
Detection	None (open-loop, passive)	Q-factor drop in resonant loop (closed-loop, active)
Operating temperature	> 1 800 °C	Ambient – 200 °C

Aspect	Heliotherm X ReGen	EMBC
Healing chemistry	Glass-forming oxide flow (SiO_2)	Si–O–Si condensation / hydrolysis cycle
Cycle limit	Unlimited (R ≈ C equilibrium)	Unlimited (condensation ↔ hydrolysis reversibility)

The EMBC system is the **low-temperature, closed-loop** counterpart to Heliotherm's **high-temperature, open-loop** model.

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2 — Identification of the EMBC Healing Phase

2.1 Which component heals?

In the YCore formula:

$$Y_{\text{Core}} = \left[n\, \mathrm{C_{12}H_{14}O_5Si(F_{\mathit{x}})} + 2\, \mathrm{XeN} + m\, (\mathrm{SiR_2O}) \right] + [\mathrm{BNNT}] + [\mathrm{GaN}]$$

three constituents have distinct mechanical roles:

Constituent	Role	Healing Function
$\mathrm{C_{12}H_{14}O_5Si(F_{\mathit{x}})}$	Structural organosilicate matrix (rigid)	Crack host — does not heal itself
$\mathrm{SiR_2O}$ (siloxane, $m \approx 24$)	Mobile polymer phase distributed in matrix	**Primary healing agent** — flows into cracks
BNNT	Anisotropic reinforcement / thermal pathway	**Crack-bridging fibre** — limits crack opening
GaN	Active electronic layer	Not involved in healing
XeN	Functional modifier	Not directly involved in healing

The siloxane phase $\mathrm{SiR_2O}$ is the EMBC analog of the Heliotherm ReGen-Phase.

2.2 Why siloxane works at low temperature

Poly(dimethylsiloxane) and related siloxanes have:

- **Glass transition temperature** $T_g \approx -125\ ^\circ\mathrm{C}$ — chains are mobile at all EMBC operating temperatures.
- **Low viscosity** at ambient conditions — siloxane oligomers can flow into micro-scale gaps via capillary action.
- **Low surface energy** ($\gamma \approx 20$–$25\ \mathrm{mN/m}$) — excellent wetting of oxide and silicate crack surfaces.
- **Reactive terminal groups** ($\mathrm{Si–OH}$, $\mathrm{Si–OR}$) — capable of forming new $\mathrm{Si–O–Si}$ bonds across crack faces.

The fluorinated organosilicate host $\mathrm{C_{12}H_{14}O_5Si(F_{\mathit{x}})}$ further assists by lowering the surface energy of crack walls (fluorine effect), enhancing capillary-driven siloxane infiltration.

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3 — The Physical Healing Mechanism (Step by Step)

> **Note:** This section describes a conceptual mechanistic pathway, not a fabrication procedure.

3.1 Phase 0 — Healthy State

The organosilicate matrix is intact. The siloxane phase is dispersed as nano-scale domains or interpenetrating channels within the rigid matrix. The BNNT network forms a connected percolation cluster. The resonator operates at:

$$Q_0 = \frac{\omega_0 \varepsilon_0 \varepsilon_{\text{eff}}}{\sigma_{\text{eff}}}$$

with $|G(\omega_0)| \geq 1$ (Barkhausen condition satisfied). The feedback loop is in steady-state oscillation.

3.2 Phase I — Damage Event (Micro-Crack Nucleation)

A micro-crack of width Δx and length ℓ nucleates in the organosilicate matrix (from thermal cycling, mechanical stress, or fatigue).

Immediate electrical consequences:

1. **Local permittivity drop:** The crack introduces an air/vacuum gap ($\varepsilon_{\text{crack}} \approx 1$) where $\varepsilon_{\text{matrix}} \approx 6$. The effective permittivity shifts:

$$\Delta\varepsilon_{\text{eff}} \approx -f_{\text{crack}} \cdot (\varepsilon_{\text{matrix}} - 1)$$

where $f_{\text{crack}} = \frac{\Delta x \cdot \ell \cdot w}{V_{\text{cell}}}$ is the crack volume fraction.

2. **Local conductivity disruption:** Current pathways through the matrix are severed at the crack face. Effective conductivity drops:

$$\sigma_{\text{eff}}(t) = \sigma_0 \cdot (1 - D_e(t))$$

where $D_e(t) \in [0, 1]$ is the **electrical damage parameter** (fractional conductance loss).

3. **Q-factor degradation:**

$$Q(t) = \frac{\omega_0 \varepsilon_0 \varepsilon_{\text{eff}}(t)}{\sigma_{\text{eff}}(t)} \quad \Rightarrow \quad Q \downarrow \text{ as } \varepsilon_{\text{eff}} \downarrow \text{ and/or } \sigma_{\text{eff}} \downarrow$$

4. **Feedback loop perturbation:** $|G(\omega_0)|$ decreases toward (but remains near) unity. The oscillation amplitude drops, signalling damage.

3.3 Phase II — Self-Activation (EM Field Concentration at Crack Tip)

This is the critical mechanistic step that distinguishes EMBC from Heliotherm.

In Heliotherm: Healing requires an *external* energy source (ambient temperature > 1 800 °C).

In EMBC: The resonator's own electromagnetic field provides the activation energy — **the system heals itself from within.**

The mechanism:

1. **Field concentration at crack tip:** The crack is a dielectric discontinuity. Maxwell's boundary conditions force the electric field to concentrate at the crack tip, where the geometry creates a singularity-like field enhancement:

$$|\mathbf{E}_{\text{tip}}| \approx |\mathbf{E}_0| \cdot \left(\frac{a}{\Delta x}\right)^{1/2}$$

where a is a characteristic material length and Δx the crack opening. Narrower cracks produce stronger field concentration.

2. **Local dielectric heating:** The concentrated field drives dielectric loss dissipation at the crack tip region:

$$P_{\text{local}} = \sigma_{\text{eff,local}} \cdot |\mathbf{E}_{\text{tip}}|^2 = \omega \varepsilon_0 \varepsilon''_{\text{eff}} \cdot |\mathbf{E}_{\text{tip}}|^2$$

This produces a **localized temperature rise** ΔT_{tip} precisely at the damage site — not throughout the bulk material.

3. **Activation of siloxane mobility:** Even a modest ΔT_{tip} of 10–50 °C above ambient significantly reduces siloxane viscosity (Arrhenius relationship):

$$\eta(T) = \eta_0 \cdot \exp\left(\frac{E_\eta}{RT}\right)$$

Since siloxane is already above T_g at ambient, the additional heating accelerates flow into the crack volume.

Result: The damage itself creates the conditions for repair — a self-activating negative feedback loop.

3.4 Phase III — Crack Infiltration and Bond Formation

With the siloxane phase mobilised at the crack tip:

1. **Capillary-driven flow:** Siloxane flows from the surrounding matrix into the crack, driven by capillary pressure:

$$P_{\text{cap}} = \frac{2\gamma \cos\theta_c}{\Delta x}$$

where γ is the siloxane surface tension, θ_c the contact angle on the organosilicate crack wall, and Δx the crack width. Narrower cracks produce *higher* capillary pressure — self-regulating.

2. **Condensation reaction across crack faces:** Reactive silanol groups on the siloxane chains and on the freshly exposed organosilicate crack surfaces undergo condensation:

$$\equiv \text{Si–OH} + \text{HO–Si} \equiv \xrightarrow{\Delta T_{\text{tip}}} \equiv \text{Si–O–Si} \equiv + \text{H}_2\text{O}$$

This forms new covalent Si–O–Si bonds that bridge the crack faces, restoring both mechanical continuity and dielectric/conductive pathways.

3. **Dielectric restoration:** The crack volume, previously filled with air ($\varepsilon \approx 1$), is now filled with siloxane ($\varepsilon_{\text{SiR}_2\text{O}} \approx 2.5\text{–}3.5$). While not identical to the original matrix permittivity ($\varepsilon \approx 6$), the effective permittivity recovers substantially:

$$\varepsilon_{\text{healed}} \approx \varepsilon_{\text{SiR}_2\text{O}} \gg \varepsilon_{\text{air}} = 1$$

3.5 Phase IV — Electrical Recovery and Loop Restabilisation

As the crack fills:

1. $\varepsilon_{\text{eff}}(t) \to \varepsilon_{\text{healed}}$ — permittivity recovers.
2. $\sigma_{\text{eff}}(t) \to \sigma_0(1 - D_{e,\text{residual}})$ — conductivity recovers (target: $D_{e,\text{residual}} \leq 0.20$, i.e., ≥ 80 % recovery).
3. $Q(t) \to Q_{\text{healed}} \lesssim Q_0$ — Q-factor recovers.
4. $|G(\omega_0)| \to |G_0|$ — Barkhausen condition re-established, stable oscillation resumes.

The feedback loop closes: damage → Q drop → field concentration → heating → siloxane flow → bond formation → Q recovery → stable state.

3.6 Phase V — Healing Phase Regeneration (Unlimited Cyclability)

This is where the Heliotherm equilibrium principle transfers directly.

The condensation reaction is **thermodynamically reversible** via hydrolysis:

$$\equiv \text{Si–O–Si} \equiv \; + \text{H}_2\text{O} \xrightarrow{\text{stress, local T}} \equiv \text{Si–OH} + \text{HO–Si} \equiv$$

Under renewed mechanical stress or at a new crack site:

- The H_2O byproduct (trapped in the sealed module) hydrolyses siloxane bonds.
- This regenerates Si–OH reactive groups — the healing precursors.
- The regenerated precursors are available for the next condensation cycle.

The siloxane phase is not consumed — it oscillates between condensed (healed) and hydrolysed (reactive) states. This is the EMBC equivalent of Heliotherm's $R(T) \approx C(T, D)$ equilibrium, yielding practically unlimited healing cycles.

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4 — Adapted ODE System for EMBC Self-Healing

Translating the Heliotherm ODE framework into the EMBC electro-chemical domain:

4.1 Electrical Damage Parameter

Replace structural damage D with **electrical damage** D_e:

$$D_e(t) = \frac{\sigma_0 - \sigma_{\text{eff}}(t)}{\sigma_0} \in [0, 1]$$

$D_e = 0$: fully healthy; $D_e = 1$: fully severed.

4.2 Damage Evolution

$$D_e(t) = D_{e,0} - H_e(E, T, t)$$

where $D_{e,0}$ is the initial electrical damage from the crack event and H_e is the electrical healing function.

4.3 Electrical Healing Function

$$H_e(E, T, t) = k_h \cdot S_{\text{SiOH}}(t) \cdot \left(1 - e^{-t/\tau_h}\right)$$

Symbol	Meaning	Unit
k_h	Healing rate constant (condensation kinetics)	s^{-1}
$S_{\text{SiOH}}(t)$	Available Si–OH concentration (normalised)	—
τ_h	Healing time constant (capillary + reaction)	s

The $(1 - e^{-t/\tau_h})$ form replaces Heliotherm's e^{-mt} to model **saturation** healing: rapid initial recovery that asymptotically approaches a maximum.

4.4 Healing Substance Dynamics

$$\frac{dS_{\text{SiOH}}}{dt} = R_{\text{hyd}}(T, [\text{H}_2\text{O}]) - C_{\text{cond}}(T, D_e(t))$$

Regeneration (hydrolysis of previously condensed bonds):

$$R_{\text{hyd}}(T, [\text{H}_2\text{O}]) = k_r \cdot [\text{H}_2\text{O}]_{\text{local}} \cdot \exp\left(-\frac{E_r}{RT}\right)$$

Consumption (condensation at active crack sites):

$$C_{\text{cond}}(T, D_e) = k_c \cdot D_e(t) \cdot \exp\left(-\frac{E_c}{RT}\right)$$

4.5 EM-Driven Activation Temperature

The local temperature at the crack tip is not an external parameter but a function of the resonator's own field:

$$T_{\text{tip}}(t) = T_{\text{ambient}} + \frac{\omega \varepsilon_0 \varepsilon''_{\text{eff}} \cdot |\mathbf{E}_{\text{tip}}|^2}{\rho\, c_p} \cdot t_{\text{dwell}}$$

where ρ is the local density, c_p the specific heat, and t_{dwell} the residence time of the concentrated field.

This couples the ODE system to the EM feedback loop: T_{tip} depends on $|\mathbf{E}_{\text{tip}}|$, which depends on crack geometry, which depends on $D_e(t)$ — a fully coupled electro-thermo-chemical system.

4.6 Equilibrium Condition (Unlimited Cyclability)

At steady state between healing events:

$$R_{\text{hyd}} \approx C_{\text{cond}} \quad \Rightarrow \quad S_{\text{SiOH}}(t) = S_{\text{stable}}$$

Exactly as in Heliotherm. The healing substance concentration equilibrates at a stable reserve level, ensuring the next damage event finds sufficient reactive Si–OH groups available.

4.7 Recovery Target

The test criterion from the Manufacturability Roadmap:

$$1 - D_e(t = 60\text{ s}) \geq 0.80 \quad \Leftrightarrow \quad \frac{\sigma_{\text{eff}}(60\text{ s})}{\sigma_0} \geq 0.80$$

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5 — BNNT Crack-Bridging: The Mechanical Limiter

The BNNT network plays a distinct but essential role: **it does not heal the crack — it limits its opening to healable dimensions.**

5.1 Bridging Condition

A BNNT fibre of length L_{BNNT} aligned at angle θ to the crack normal can bridge the crack if:

$$\Delta x_{\text{crack}} < L_{\text{BNNT}} \cdot \cos\theta$$

For aligned BNNTs (Herman's parameter ≥ 0.6 per Stage A gate), the effective bridging length is:

$$L_{\text{bridge}} \approx L_{\text{BNNT}} \cdot \langle \cos\theta \rangle \approx 0.8 \cdot L_{\text{BNNT}}$$

5.2 Why BNNT holds

BNNT tensile strength is approximately 30 GPa — among the highest of any known material. The nanotubes do not fracture at the crack face. Instead, they:

1. **Bridge the crack mechanically** — holding crack faces within $\Delta x < L_{\text{bridge}}$.
2. **Maintain thermal pathways** across the crack — preventing thermal runaway from dielectric heating.
3. **Provide nucleation sites** for siloxane condensation — the BN surface chemistry promotes Si–OH attachment.

5.3 Percolation Recovery

If the BNNT network is above the percolation threshold ($f_{\mathrm{BNNT}} > f_c$, typically $f_c \approx 0.03$–0.10 for high-aspect-ratio tubes), a single crack does not sever the connected cluster. The percolation backbone reroutes around the damage zone, maintaining global thermal and (if applicable) electrical connectivity while local healing proceeds.

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6 — Closed-Loop vs. Open-Loop: The EMBC Advantage

6.1 Comparison of Healing Architectures

Eigenschaft / Merkmal	HELIOTHERM X ReGen (Open-Loop)	EMBC (Closed-Loop)
Grundprinzip	Passiv (Passive)	Aktiv (Active)
Prozessablauf	Damage ➔ ReGen flows ➔ Crack fills ➔ Done	Damage ➔ Q drops ➔$$
Aktivierungsbedingung	Benötigt externe Hitze (T > 1800 °C)	Selbstaktivierend (Self-activating) durch lokales Feld
Detektion	Keine Erkennung (No detection)	Selbstdetektierend (Self-detecting) durch Abfall von Q
Feedback-Mechanismus	Kein Feedback (No feedback)	Geschlossene Feedback-Schleife (Closed-Loop)
Limitierung / Dauer	Heilt nur im extern erhitzten Zustand (heals only while hot)	Selbstlimitierend (Self-limiting), Prozess stoppt, sobald Q stabilisiert ist

6.2 Two Operating Regimes

Regime	Condition	Mechanism	Speed
Resonant (Active)	Resonator powered, $\|G\| \geq 1$	EM field drives healing via dielectric loss at crack tip	Fast (seconds)
Passive (Quiescent)	Resonator unpowered or $\|G\| < 1$	Capillary-driven siloxane flow only (no EM activation)	Slow (minutes–hours)

The resonant regime is the primary healing mode. The passive regime provides background healing when the resonator is not energised, albeit at a reduced rate — analogous to Heliotherm operating below 1 800 °C where healing slows but does not stop entirely if residual thermal energy is present.

7 — Coupled System: Complete Equation Set

For reference, the full coupled EMBC self-healing system in one block:

$$D_e(t) = D_{e,0} - k_h \cdot S_{\mathrm{SiOH}}(t) \cdot (1 - e^{-t/\tau_h})$$

$$\frac{dS_{\mathrm{SiOH}}}{dt} = k_r \cdot [\mathrm{H_2O}] \cdot e^{-E_r/RT_{\mathrm{tip}}} - k_c \cdot D_e(t) \cdot e^{-E_c/RT_{\mathrm{tip}}}$$

$$T_{\mathrm{tip}}(t) = T_0 + \frac{\omega_0 \varepsilon_0 \varepsilon''_{\mathrm{eff}} \cdot |\mathbf{E}_0|^2}{4\rho c_p} \cdot \frac{a}{\Delta x(t)}$$

$$Q(t) = \frac{\omega_0 \varepsilon_0 \varepsilon'_{\mathrm{eff}}(t)}{\sigma_0 (1 - D_e(t))}$$

$$|G(\omega_0, t)| = \frac{K_{\mathrm{mat}} \cdot K_{\mathrm{fb}}}{1 + Q(t)^{-2} \cdot \delta\omega^2/\omega_0^2} \geq 1$$

$$\Delta x(t) \leq L_{\mathrm{BNNT}} \cdot \langle \cos\theta \rangle \quad \text{(BNNT bridging constraint)}$$

Initial conditions: $D_e(0) = D_{e,0}$, $S_{\mathrm{SiOH}}(0) = S_{\mathrm{stable}}$, $\Delta x(0) = \Delta x_0$.

Boundary condition (recovery target): $D_e(60\ \mathrm{s}) \leq 0.20$.

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8 — Illustrative Parameter Table

> All values are assumed / literature-plausible and require experimental validation.

Parameter	Symbol	Value (assumed)	Unit	Source / Status
Healing rate constant	k_h	0.05 – 0.15	s^{-1}	to be measured
Healing time constant	τ_h	5 – 20	s	to be measured
Hydrolysis activation energy	E_r	40 – 60	kJ/mol	literature (siloxane)
Condensation activation energy	E_c	30 – 50	kJ/mol	literature (siloxane)
Siloxane surface tension	γ	20 – 25	mN/m	literature (PDMS)
Siloxane viscosity (25 °C)	η_0	0.5 – 50	Pa·s	depends on MW
BNNT tensile strength	σ_{BNNT}	~30	GPa	literature
BNNT typical length	L_{BNNT}	1 – 10	μm	supplier-dependent
Critical crack width (healable)	$\Delta x_{\max}$	< 1	μm	$L_{\mathrm{BNNT}} \cdot \langle \cos\theta \rangle$
Crack-tip field enhancement factor	$a/\Delta x$	10 – 100	—	geometry-dependent
Local temperature rise at tip	ΔT_{tip}	10 – 50	°C	to be simulated
Water content (sealed module)	$[\mathrm{H_2O}]$	trace (ppm-level)	—	byproduct of condensation

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9 — Validation Checklist (Experimental)

#	Test	Method	Pass Criterion
V-1	Siloxane flow into artificial micro-crack	SEM cross-section after controlled indent	Crack volume ≥ 70 % filled within 60 s
V-2	Si–O–Si bond formation across crack	FTIR / Raman on healed region	Characteristic Si–O–Si stretch at ~1 050 cm^{-1}
V-3	Conductance recovery (4-point probe)	In-situ resistance measurement pre/post crack	$\sigma_{\text{eff}}/\sigma_0 \geq 0.80$ at $t = 60$ s
V-4	Dielectric recovery (impedance analyser)	ε_{eff} measurement at 13.56 MHz pre/post crack	$\varepsilon_{\text{healed}}/\varepsilon_0 \geq 0.70$
V-5	Healing cyclability	Repeated indent + 4-point probe (≥ 10 cycles)	No progressive degradation of recovery ratio
V-6	H_2O detection in sealed module	Karl Fischer titration or mass spec on outgas	$[\text{H}_2\text{O}]$ present at ppm level, confirms condensation pathway
V-7	EM field concentration at crack (simulation)	FDTD or FEM of cracked geometry at 13.56 MHz	$\lVert\mathbf{E}_{\text{tip}}\rVert/\lVert\mathbf{E}_0\rVert \geq 5$
V-8	BNNT bridging (in-situ)	SEM of crack with visible nanotube bridges	BNNTs spanning crack at $\Delta x < L_{\text{bridge}}$
V-9	Passive healing rate	Same as V-3 but with resonator unpowered	σ recovery ≥ 50 % within 600 s (10 min)
V-10	Q-factor recovery (VNA)	S_{11} measurement before and after healing	$Q_{\text{healed}}/Q_0 \geq 0.75$

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10 — Summary: How EMBC Self-Healing Works (One-Paragraph Mechanistic Statement)

When a micro-crack nucleates in the EMBC organosilicate matrix, the local dielectric discontinuity concentrates the resonator's own 13.56 MHz electromagnetic field at the crack tip, producing localised dielectric heating. This thermal activation mobilises the siloxane (SiR_2O) healing phase — already above its glass transition at ambient temperature — which flows into the crack via capillary action. Reactive silanol groups (Si–OH) on the siloxane chains and on the freshly exposed crack surfaces undergo condensation ($\text{Si–OH} + \text{HO–Si} \rightarrow \text{Si–O–Si} + \text{H}_2\text{O}$), forming new covalent bonds that bridge the crack and restore both dielectric permittivity and electrical conductivity. The BNNT network limits crack opening to dimensions small enough for siloxane infiltration and provides mechanical bridging and thermal management. The water byproduct, trapped in the sealed module, regenerates Si–OH groups via hydrolysis at future damage sites, making the healing phase self-renewing and the cycle count practically unlimited — the low-temperature, closed-loop analog of the Heliotherm X ReGen equilibrium principle. The Q-factor of the resonant loop serves simultaneously as the damage sensor (Q drops on cracking) and the healing driver (Q depression concentrates the activating field), closing the feedback loop autonomously without external intervention.

Disclaimer & Liability Waiver

Please be advised that the methodology, models, and architecture outlined herein represent only one possible approach to exploring this concept. Because this project ventures into absolute uncharted technological territory, we can offer no guarantees regarding its functional success, physical manufacturability, or real-world viability.

This framework is intended to serve strictly as a theoretical foundation and must be evaluated and applied with extreme caution and rigorous critical thinking. We explicitly assume no liability or responsibility for any outcomes, risks, damages, or failures that may arise from the application, adaptation, or experimental testing of the information provided.

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Structural analog: Heliotherm X ReGen (Self-Healing Mechanism, Section 5)